# UML

## (Unified Modeling Language)

Smghega, ho skta h thoda complex lge but last ma smghege ki actual interviews ma kisko dhyan rkhna ha
(interviews ma bohot hi rough aur basic version bnate h iska
not everything)

UML

- Structural (static) — what exisits in the system? — class Diagram
- Behavioral (dynamic) — what happens on runtime? — sequence diagram

## Class Diagram

structural diagram that shows the **classes** in your system,
their attributes, methods, and the **relationships** between them.

**blueprint of your code**

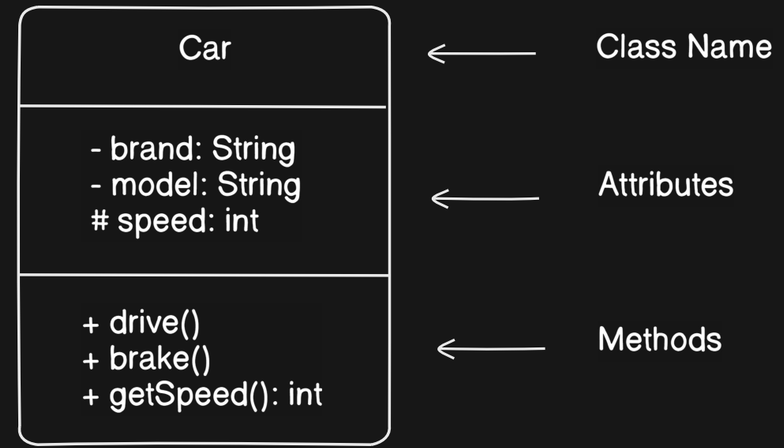
## Class Diagram

Class Structure

Relationships

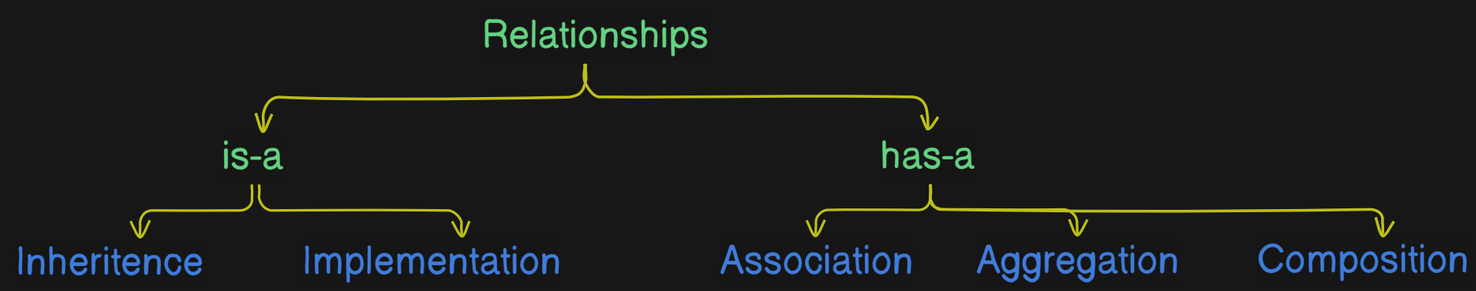
## Class Structure

```java
class Car {
    private String brand;
    private String model;
    protected int speed;

    public void drive() { }
    public void brake() { }
    public int getSpeed() { }
}
```




| | |
|---|---|
| private | - |
| protected | # |
| public | + |

## Relationships:



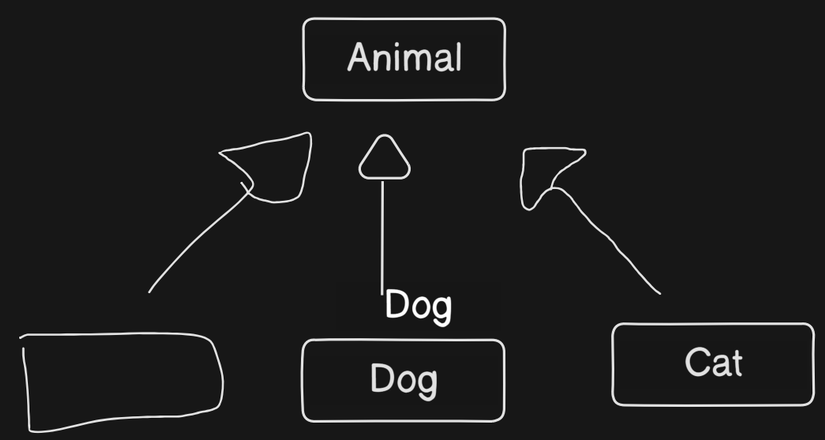
is-a

- Inheritence
- Implementation

## Inheritence

B extends A

```
class Animal {
  public void run(){}
}

class Dog extends Animal { }

class Cat extends Animal { }
```

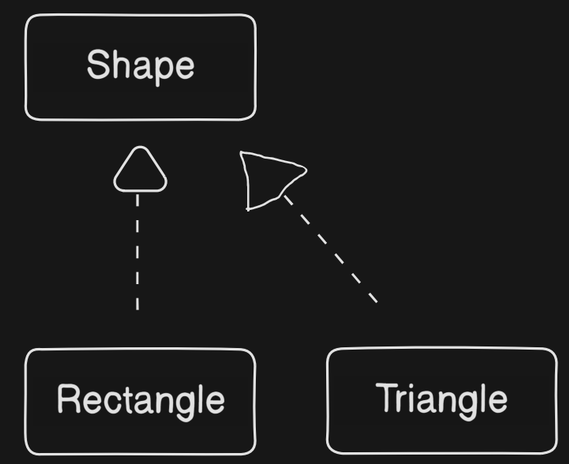



Dog is a Animal

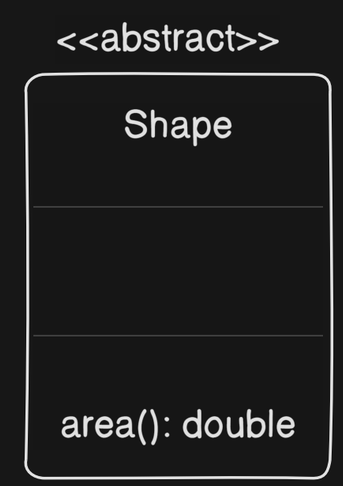
## Implementation:

B implements A

```
interface Shape {
    double area();
}

class Rectangle implements Shape {
    public double area() { }
}

class Triangle implements Shape {
    public double area() { }
}
```

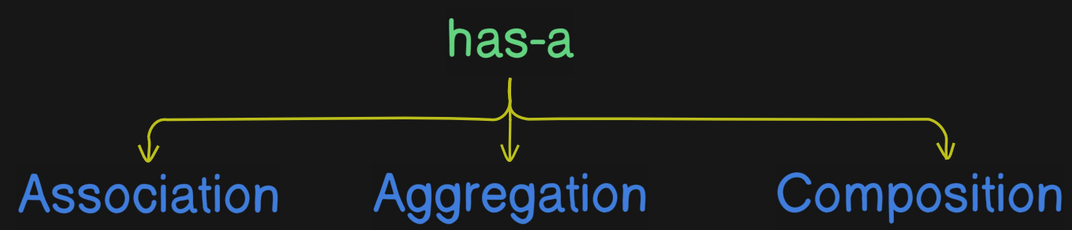



Rectangle is a shape
Triangle is a shape

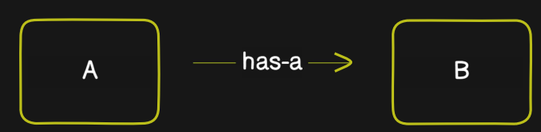



is-a

Inheritence Implementation



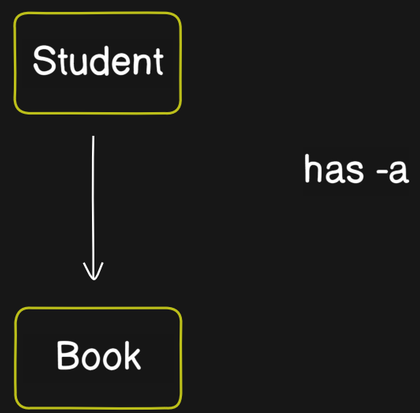

```
class A {
    private B b;
    A(B obj){}

}

class B {

}
```




A has a B

## Association:

(knows about)



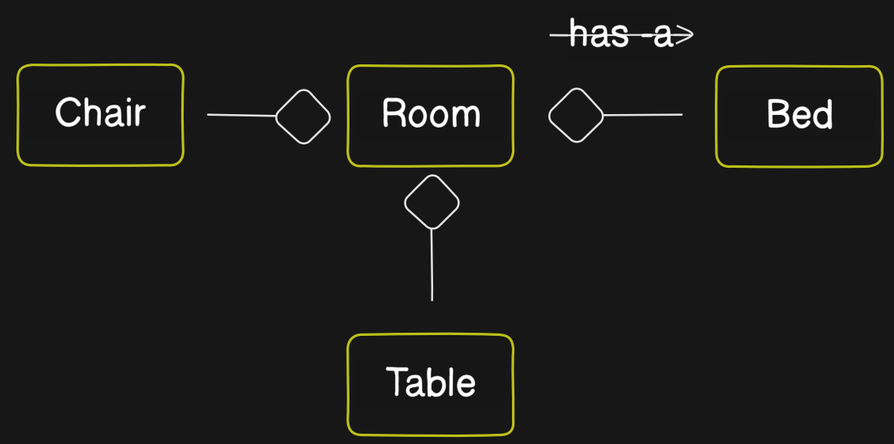

has -a

```
class Student{
private Book b;
}

class Book{}
```

## Aggregation

(has-a, loosely)

Room has bed
Room has table
Room has chair

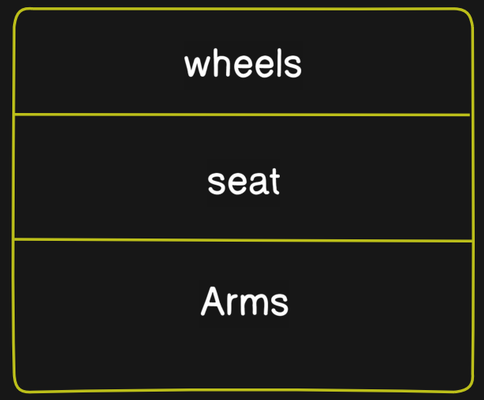


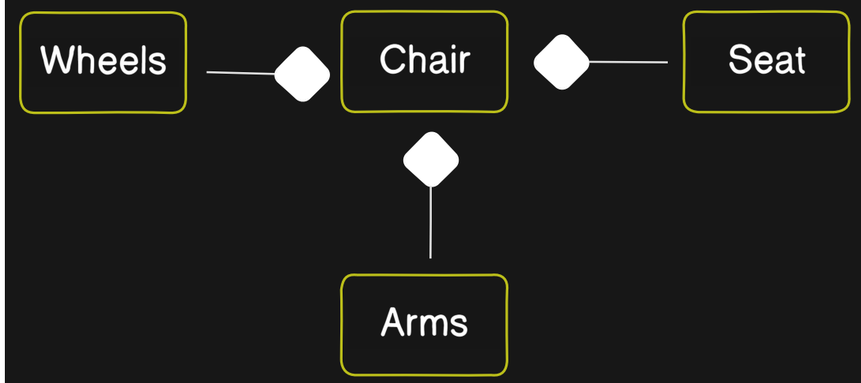

contained objects can exist independently

## Composition

(has-a, tightly)

strongest




contained objects cant exist without parent

Associations - knows about

Aggregation - contains, individual existence

Composition - contains, made up of, no individual existence of contained objects

composition

inheritance

implementation

inheritence

- UML Class Diagrams
- Class Structure
- Relationships

  is-a

  has-a

How to make it quickly?